

Labwork 6 - Map

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1 GPU Map Implementations

In this labwork, we will learn how to perform mapping functions inside the CUDA kernel. Three image processing operations are implemented using 2D CUDA grids. The same 718×718 image as in the previous labwork is used. A second image of the same size is used for blending.



Figure 1: Input images (718×718).

For binarization, the RGB values of each pixel are first averaged to obtain a grayscale intensity. A threshold of 128 is then applied: pixels below the threshold are set to 0, while the others are set to 255. The resulting value is written to all three RGB channels.



Figure 2: Binary image.

For brightness control, a constant value of 100 is added to or subtracted from each RGB channel to produce brighter and darker images, respectively. The resulting values are clipped to the range $[0, 255]$ to prevent overflow and underflow.



Figure 3: Brightness control images.

For image blending, corresponding pixels from two images are combined using

$$Q(x, y) = c\Phi_1(x, y) + (1 - c)\Phi_2(x, y),$$

In this experiments, c value is varied, where lower c value result in a image that is more similar to the second image, while higher c value result in a image that is more similar to the first image.



Figure 4: Blending images. a) $c = 0.2$, b) $c = 0.5$, c) $c = 0.8$.

2 Results

The GPU implementations were measured over 10 runs for each 2D block size. A warm-up run was performed before timing. The measured time includes transferring image data to the GPU and copying the result back to the host.

Table 1: Average GPU execution time (ms $\pm$ std) for different 2D block sizes.

Block size	Binarization	Brightness	Blending
8×8	2.2 ± 0.6	2.5 ± 0.8	3.7 ± 0.9
8×16	2.3 ± 0.6	2.0 ± 0.5	3.4 ± 0.7
16×8	2.3 ± 0.6	2.6 ± 0.9	3.4 ± 0.7
8×32	2.6 ± 0.9	2.3 ± 0.7	3.4 ± 0.7
32×8	2.3 ± 0.6	2.3 ± 0.6	3.5 ± 0.7
16×16	2.3 ± 0.6	2.2 ± 0.6	3.4 ± 0.7
32×32	2.4 ± 0.6	2.3 ± 0.6	3.6 ± 0.9
16×32	2.3 ± 0.6	2.2 ± 0.6	3.4 ± 0.7
32×16	2.3 ± 0.6	2.5 ± 0.9	3.5 ± 0.8

Binarization and brightness control have comparable execution times across the tested block sizes, with their average times generally falling within a similar range at 2.0–2.6 (ms). Blending takes longer, with execution times of 3.4–3.7 (ms), as each thread needs to access two input images instead of one.

The execution time remains relatively stable across different block sizes, with no consistent improvement as the block size increases. Binarization is

fastest with 8×8 blocks (2.2 ms), while brightness control is fastest with 8×16 blocks (2.0 ms). For blending, several configurations achieve the lowest average time of 3.4 ms.

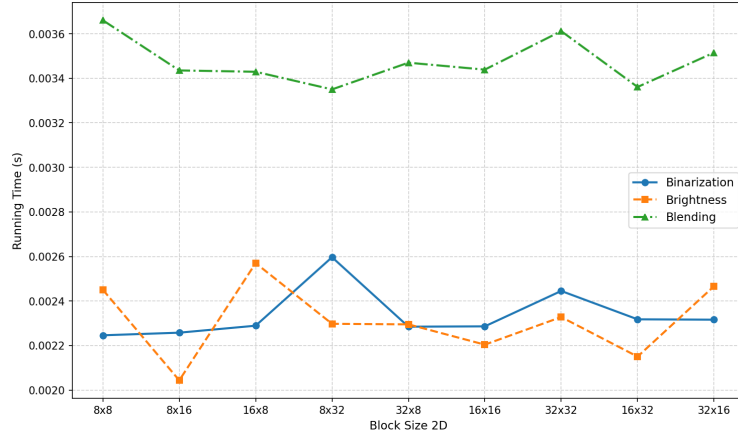


Figure 5: GPU execution time for different 2D block sizes.

2.1 Conclusion

Across the tested configurations, execution times remain relatively stable, with no single block size providing the lowest time for all operations.